JAVA PROGRAM REPORT

Interface Example

Programming Language:	Java
Topic:	Interface Implementation
Program File:	InterfaceExample.java

Objective

To demonstrate interface implementation in Java, where the UPI class implements the pay() method declared in the Payment interface.

Java Source Code

```
interface Payment {
    void pay();
}

class UPI implements Payment {

    public void pay() {
        System.out.println("Payment made through UPI");
    }
}

public class InterfaceExample {
    public static void main(String[] args) {

        UPI u = new UPI();
        u.pay();
    }
}
```

Program Explanation

1. Payment Interface: Declares the pay() method without providing its implementation.

2. UPI Class: Implements the Payment interface and defines the pay() method.

3. Method Implementation: The pay() method prints “Payment made through UPI”.

4. Main Method: Creates a UPI object and calls u.pay().

Expected Output

```
Payment made through UPI
```

Execution and Output Screenshots

Test Case 1: Compilation and Execution

```
C:\Users\Student> javac InterfaceExample.java

C:\Users\Student> java InterfaceExample

Payment made through UPI

C:\Users\Student>
```

Verification: The program compiles and displays the message implemented by the UPI class.

Test Case 2: Re-execution Verification

```
C:\Users\Student> java InterfaceExample

Payment made through UPI

C:\Users\Student>
```

Verification: Re-running the class produces the same output.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	Compilation and execution succeed.	Payment message is displayed.	PASS
Test Case 2	Same output on re-execution.	Output remains consistent.	PASS

Conclusion

The program demonstrates Java interface implementation. The Payment interface declares the pay() method, and the UPI class provides its implementation. The output confirms that the method executes as expected.